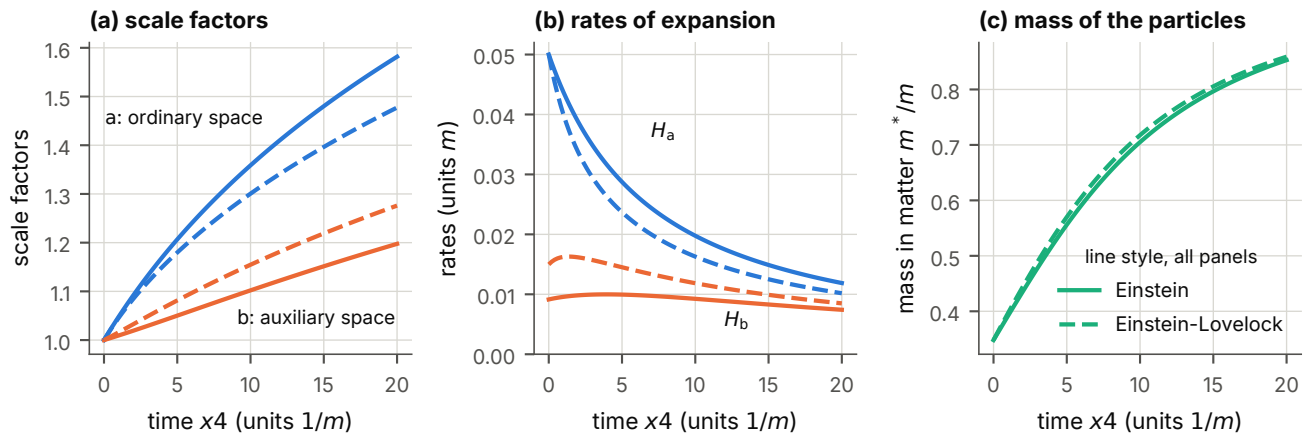
(a) scale factors
scale factors
1.6
1.5
1.4
1.3
1.2
1.1
1.0
a: ordinary space
b: auxiliary space
0
5
10
15
20
time x4 (units 1/m)
(b) rates of expansion
rates (units m)
0.05
0.04
0.03
0.02
0.01
0.00
H_a
H_b
0
5
10
15
20
time x4 (units 1/m)
(c) mass of the particles
mass in matter m*/m
0.8
0.7
0.6
0.5
0.4
line style, all panels
Einstein
Einstein-Lovelock
0
5
10
15
20
time x4 (units 1/m)